

Proton Pressure in Terms of Degenerate Quark Distribution

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Most of the ordinary matter in the universe is composed of protons, which in turn consist of quarks and gluons. Proton is a stable microscopic platform, allowing all four forces – strong, electromagnetic, weak and gravity – to act in terms of their respective form factors, where the gravitational form factor describes the hadron's mass, spin, and D-term [1]. In 2018, the D-term was extracted experimentally via Deeply Virtual Compton Scattering, and the pressure distribution inside the proton was sketched [2]. The measurement revealed a strongly repulsive force within 0 - 0.6 fm, and its peak value 10^{35} Pa is noticed around 0.2 fm. This peak value is much larger than the neutron star pressure. A similar distribution was also predicted from the Lattice quantum chromodynamics (LQCD) calculations [3], which also provided the shear distribution pattern.

The present work is aimed to map the pressure distribution at zero temperature or degenerate pressure of quark inside the proton, where their chemical potential or Fermi energy is considered as a radial-dependent tuning parameter to obtain the pressure distribution, predicted by Ref. [2]. From a physics point of view, a complicated three-body quark potential, along with surrounding sea quarks and gluons, is responsible for this mechanical pressure distribution. In the quantum field theoretical picture, this can be resolved using non-perturbative QCD calculations within the proton structure. However, this is currently missing in the literature and may be explored soon. The radial-dependent chem-

ical potential in the present work effectively maps this interaction measurement within the proton structure. This attempt is very simple but may be very useful to deal with different quantitative distributions inside protons, which can be compared with other well-studied systems like the neutron star (NS), quark-gluon plasma (QGP), etc.

In Large Hadron Collider (LHC) experiments, a super hot QGP system is expected for nucleus-nucleus collisions and even proton-proton collisions.

This system can be described by basic statistical physics, assuming the quark follows a Fermi-Dirac (FD) distribution function:

$$f(\epsilon) = \frac{1}{e^{(\epsilon-\mu)/T} + 1} \quad (1)$$

where $\epsilon = \sqrt{p^2 + m^2}$, μ and T are the energy, chemical potential, and temperature, respectively. For the LHC case, we expect a zero net quark density state with $\mu = 0$ and an ultra-relativistic quark with $m = 0$. The quark pressure can then be obtained as

$$\begin{aligned} P(r) &= \frac{g}{h^3} \int_0^\infty \left(\frac{pv}{3}\right) f(\epsilon) d^3p \\ &= \frac{7}{8} \frac{g}{\pi^2} \zeta(4) T^4. \end{aligned} \quad (2)$$

On the other side of the QCD Phase diagram, we expect degenerate quark matter inside the core of the neutron star, where FD distribution in the limit $T \ll \mu$ modifies to the simple form:

$$f(\epsilon) = \begin{cases} 0 & \epsilon > \mu \\ 1 & \epsilon < \mu \end{cases} \quad (3)$$

And the pressure has the following form:

$$P = \frac{g}{24\pi^2} \mu^4. \quad (4)$$

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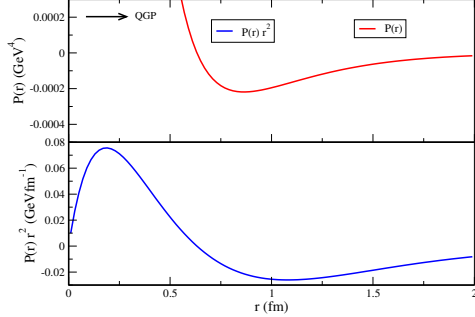


FIG. 1: The pressure (upper panel) and force (lower panel) distribution inside the proton. Arrow indicating approximated order of magnitude for QGP system.

Now, in these two extreme cases – QGP in LHC (at $\mu = 0$, $T \neq 0$) and quark matter inside NS (at $\mu \neq 0$, $T = 0$) – all thermodynamical quantities, like pressure, energy density, etc., have a simple T^n and μ^n dependence. Now, this $P \propto \mu^4$ expression of dense quark matter in NS core may be adopted to describe the proton pressure, where we may consider μ as an indirect interaction measurement instead of an actual thermodynamical parameter.

To do this mapping, we first found out a parametric function for the pressure distribution, provided by Ref. [2]:

$$P(r)r^2 = 25.93 \times 10^{-2} (4r - 6.325r^2)e^{-6.325r/2} \quad (5)$$

and then we tuned our μ to fit the pressure.

Using the parametric function of Eq. 5, we have plotted the radial proton pressure distribution $P(r)$ (upper panel) and force distribution $P(r)r^2$ (lower panel) in Fig. 1. The force distribution shows a strong repulsion up to about 0.6 fm and attraction thereafter. At about $r = 0.17$ fm, the pressure reaches a peak value with $P(r)r^2 = 7.5 \times 10^{-2} \text{ GeVfm}^{-1}$. Remarkably, this value crosses the neutron star pressure and the approximated order of magnitude for QGP pressure. Table I compares the orders of magnitude for a few systems of interest, and the proton exhibits the highest pressure observed to date, for which the topic

System	Pressure (Pa)
Intergalactic void	10^{-17}
Earth sea level	10^5
Pressure inside Earth's core	10^{11}
Pressure inside Sun's core	10^{16}
White Dwarf	10^{23}
NS	10^{32}
proton	10^{35}
QGP	$10^{34}-10^{36}$

TABLE I: Orders of magnitude of pressure for a few systems, obtained from the wikipedia link – orders of magnitude.

becomes quite a matter of interest for the community.

In future, we aim to undertake more rigorous calculations of the system by examining quantities like the number density, enthalpy density, shear stress, etc. We also plan to introduce the bound state wave function of the proton by connecting our effective chemical potential. This formalism is in progress and maybe a very relevant investigation for the future experimental facility - Electron-Ion Collider (EIC) at BNL, USA.

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